THE INVERSE OF THE TRIANGLE $[\,k \le n \le 2k\,]/C_n$ HAS INTEGER ENTRIES: A PROOF OF BARRY'S CONJECTURE IN OEIS A127767

MATH0

Abstract. Let C_n be the Catalan numbers and let A be the infinite lower-triangular matrix with $A(n,k) = 1/C_n$ if $k \le n \le 2k$ and $A(n,k) = 0$ otherwise. We prove that every entry $T(n,k)$, $0 \le k \le n$, of the inverse matrix $T = A^{-1}$ is an integer; the proof shows that it is in fact an integer multiple of C_k. This settles a conjecture that Paul Barry recorded in OEIS entry A127767 in 2007 and that is still listed there as open. We classify the result as new, but the proof is elementary: it amounts to the factorization $A = \mathrm{diag}(1/C_n)\,M$ with M a unitriangular 0/1 matrix, the same idea as a 2021 formula of R. J. Mathar for a sibling triangle. The theorem is verified in Lean 4 against the frozen, fidelity-reviewed statement that math0 formalized from the OEIS entry; in that statement T is defined by forward substitution, proved to invert every finite truncation of A, and checked by the kernel against all 66 listed terms.

1. Introduction

Let $C_n = \frac{1}{n+1}\binom{2n}{n}$ be the n-th Catalan number (A000108 [5]). In January 2007 Paul Barry added to the OEIS a family of number triangles, each the inverse of a lower-triangular matrix supported on the band $k \le n \le 2k$ whose nonzero entries in row n all equal $1/d(n)$ for some positive arithmetic function d: for example $d(n) = n+1$ (A127710 [8]), the Fibonacci number F_{n+1} (A127711 [9]), $2n+1$ (A127749 [10]), $n!$ (A127755 [11]), C_n (A127767 [12]), Euler's totient $\varphi(n+1)$ (A127771 [13]) and $\lfloor (n+2)/2 \rfloor$ (A127793 [14]). For most of these he recorded the conjecture that the inverse has integer entries. This note concerns the Catalan case, A127767. Its entry is named

> `Inverse of number triangle A(n,k) = 1/C(n) if k <= n <= 2k, 0 otherwise, where C(n) = A000108(n).`

and its comment line reads, verbatim,

> `It is conjectured that all elements of the triangle are integers.`

The entry was submitted by Paul Barry on January 28, 2007. In its current revision (#11, September 30, 2018) the statement is still labelled a conjecture. The entry has no formula section; besides the data, the examples and cross-references it contains only a PARI program (Michel Marcus, 2018) that computes the rows by inverting a finite matrix. We prove the conjecture (Theorem 1), with a Lean proof of the full statement.

Prior work and status. The math0 literature review found the following related work and no prior resolution of the conjecture in A127767.

- The 0/1 triangle $M(n,k) = [\,k \le n \le 2k\,]$ is A101688 [6], read as a triangle, and its inverse is A111967 [7] (P. Barry, 2005). Since M is lower unitriangular with integer entries, A111967 is an integer triangle. The identity $T(n,k) = C_k \cdot \mathrm{A111967}(n,k)$, which we prove as Corollary 3, settles A127767; it does not appear in either entry.

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- In A127749 (the case $d(n) = 2n+1$), R. J. Mathar added on April 21, 2021 the formula $T(n,k) = (2k+1)\cdot \mathrm{A111967}(n,k)$ [10]. This is the same factorization applied to a sibling triangle, and it is the closest prior use of the method. A127749 itself carries no integrality conjecture, and the factorization was not recorded for A127767 or stated as a general principle.
- For A127793 (the case $d(n) = \lfloor (n+2)/2 \rfloor$) the integrality conjecture was proved by FUNG Cheok Yin (comment in the entry, September 2022) and by S. Fried [1], who also gave an explicit formula for the entries. Fried's paper treats only A127793 and does not state a general factorization.
- The conjectures in A127710, A127711, A127755 and A127771 are, like A127767, still listed as open in the OEIS; the argument of this note settles them as well (Remark 4). A search of arXiv, MathOverflow, Mathematics Stack Exchange and the SeqFan list found nothing on A127767.

We therefore classify Theorem 1 as a *new result*, with the caveat that it is an elementary observation rather than a substantive theorem: once A is written as $\mathrm{diag}(1/C_n)\,M$, integrality is a one-line consequence.

2. Statement

Infinite lower-triangular matrices over $\mathbb{Q}$, indexed by $\mathbb{N}\times\mathbb{N}$, form a ring: entries of a product are finite sums. A lower-triangular matrix with nonzero diagonal entries has a unique two-sided inverse in this ring, and the inverse is again lower triangular. Moreover, the $N\times N$ leading truncation of a product of lower-triangular matrices is the product of the truncations, so T is the inverse of A if and only if, for every N, the truncation T_N is the inverse of the truncation A_N.

Theorem 1. *Let A be the lower-triangular matrix with entries*

$$A(n,k) = \begin{cases} 1/C_n & \text{if } k \le n \le 2k, \\ 0 & \text{otherwise,} \end{cases} \qquad n,k \ge 0,$$

and let $T = A^{-1}$. Then $T(n,k) \in \mathbb{Z}$ for all $0 \le k \le n$.

Since T is lower triangular, $T(n,k) = 0$ for $k > n$, so every entry of T is an integer. The proof gives more: $T(n,k) \in C_k\mathbb{Z}$ for all n,k (Proposition 2).

The rows $n = 0,\dots,4$ of T are (1), $(0,1)$, $(0,-1,2)$, $(0,1,-2,5)$, $(0,0,0,-5,14)$.

Lean formalization. The statement was formalized by math0 in `lean/Math0/Statements/OEIS/A127767.lean`, reviewed for fidelity, and frozen (SHA-256 `e61b7ad2...9dce`) before any proof attempt. Its definitions and the statement read as follows. The listing is verbatim, except that we omit the import, comments, blank lines, all theorems, and the definitions `rowOf` and `a` (reading the triangle by rows) and `Amat`, `Tmat` (the $N\times N$ truncations of `A` and `T`, as matrices `Matrix (Fin N) (Fin N) ℚ`).

```
namespace Math0.Statements.A127767
open Finset
def cat : ℕ → ℕ
  | 0 => 1
  | n + 1 => cat n * (2 * (2 * n + 1)) / (n + 2)
def A (n k : ℕ) : ℚ := if k ≤ n ∧ n ≤ 2 * k then 1 / (cat n : ℚ) else 0
def nextRow (rs : List (List ℚ)) : List ℚ :=
  (List.range (rs.length + 1)).map fun k =>
    (cat rs.length : ℚ) * ((if rs.length = k then 1 else 0) -
```

```
      ∑ j ∈ range rs.length, A rs.length j * (rs.getD j []).getD k 0)
def rows : ℕ → List (List ℚ)
  | 0 => []
  | n + 1 => rows n ++ [nextRow (rows n)]
def T (n k : ℕ) : ℚ := ((rows (n + 1)).getD n []).getD k 0
def statement : Prop := ∀ n k : ℕ, k ≤ n → ∃ z : ℤ, T n k = z
end Math0.Statements.A127767
```

How the definitions encode the entry. The function `cat` computes C_n by the product recursion $C_{n+1} = C_n \cdot 2(2n+1)/(n+2)$, in which the natural-number division is exact; the file proves `cat_eq_catalan`: `cat n = catalan n` (Mathlib's Catalan numbers) and `cat_pos`: `0 < cat n`, so `1 / (cat n : ℚ)` never divides by zero. The rational number `A n k` is $A(n,k)$, exactly as in the name of the entry. The rows of T are built one at a time: if `rs` holds rows $0,\dots,n-1$ (so `rs.length` $= n$), then `nextRow rs` is the list of the numbers

$$T(n,k) = C_n\Big([\,n=k\,] - \sum_{j<n} A(n,j)\,T(j,k)\Big), \qquad 0 \le k \le n, \tag{1}$$

where `(rs.getD j []).getD k 0` reads $T(j,k)$, with value 0 when $k > j$. This is forward substitution for the equations $\sum_{j\le n} A(n,j)T(j,k) = [\,n=k\,]$, using $A(n,n) = 1/C_n$. The values are rationals; integrality is not built into the types. That `T` is the inverse of `A` is proved, not assumed. The file proves the following, with proofs omitted here:

```
theorem T_spec (n k : ℕ) :
    T n k = if k ≤ n then
      (cat n : ℚ) * ((if n = k then 1 else 0) - ∑ j ∈ range n, A n j * T j k) else 0
theorem inv_Amat (N : ℕ) : (Amat N)⁻¹ = Tmat N
theorem statement_iff_matrix :
    statement ↔ ∀ (N : ℕ) (i j : Fin N), ∃ z : ℤ, (Amat N)⁻¹ i j = z
```

Here `T_spec` is (1) together with $T(n,k) = 0$ for $k > n$, and `(Amat N)`$^{-1}$ is Mathlib's matrix inverse. The proof of `inv_Amat` shows $\sum_{x<N} A(n,x)T(x,k) = [\,n=k\,]$ for $n < N$ (`sum_A_mul_T`): the terms with $x > n$ vanish, and the term $x = n$ is $T(n,k)/C_n$, which `T_spec` identifies. Hence $A_N T_N = 1$, and a right inverse of a square matrix is its inverse (`Matrix.inv_eq_right_inv`). By the remark at the start of this section, `statement` is exactly the assertion of Theorem 1; the entries with $k > n$, which are not part of the statement, are 0 (`T_eq_zero_of_lt`). By `statement_iff_matrix` it is also equivalent to the integrality of every finite inverse A_N^{-1}.

Data checks. The file defines the triangle read by rows, `a m = T n k` for $m = n(n+1)/2+k$, $0 \le k \le n$, with offset 0 as in the entry. Two theorems proved by `decide +kernel` check that `a` reproduces all 66 terms of the OEIS data (`a_data`) and that `rows 11` is the list of rows $0,\dots,10$ (`rows_data`). Here the Lean kernel itself evaluates `cat`, `A` and `rows` with exact rational arithmetic and compares the result with the literal list. A script confirmed that this list equals the `%S`, `%T`, `%U` lines of the entry. These checks anchor the definitions to the entry independently of the conjecture; the proof below does not use them.

3. Proof

The proof follows `lean/Math0/Frontier/OEIS/A127767.lean`. It works directly with the recursion `T_spec` and does not use matrices. Write

$$M(n,j) = [\,j \le n \le 2j\,] \in \{0,1\}, \qquad \text{so that} \qquad A(n,j) = M(n,j)/C_n.$$

From the statement file we use only $C_n > 0$ (`cat_pos`) and the following form of `T_spec`: for all n, k,

$$T(n,k) = \begin{cases} C_n\big([\,n=k\,] - \sum_{j<n} A(n,j)\,T(j,k)\big) & \text{if } k \le n, \\ 0 & \text{if } k > n. \end{cases} \tag{2}$$

Proposition 2 (`T_eq_int_mul_cat`)**.** *For all $n, k \in \mathbb{N}$ there is $z \in \mathbb{Z}$ with $T(n,k) = z\,C_k$.*

Proof. By strong induction on n, for all k simultaneously. Fix n and k, and assume that for every $j < n$ there is $z_j \in \mathbb{Z}$ with $T(j,k) = z_j C_k$. (In Lean, `choose!` turns this hypothesis into a function $f\colon \mathbb{N} \to \mathbb{Z}$ with $T(j,k) = f(j)\,C_k$ for all $j < n$; we write $z_j = f(j)$.)

If $k > n$, then $T(n,k) = 0 = 0 \cdot C_k$ by (2).

Let $k \le n$. For $j < n$,

$$A(n,j)\,T(j,k) = \frac{M(n,j)}{C_n}\,z_j C_k = \frac{C_k}{C_n}\,M(n,j)\,z_j$$

(in Lean, both cases of the definition of A are closed by `push_cast` and `ring`). Summing over $j < n$,

$$\sum_{j<n} A(n,j)\,T(j,k) = \frac{C_k}{C_n}\,W, \qquad W = \sum_{j<n} M(n,j)\,z_j \in \mathbb{Z}.$$

By (2), and since $C_n \neq 0$,

$$T(n,k) = C_n[\,n=k\,] - C_k\,W.$$

If $n = k$, then $C_n[\,n=k\,] = C_k$; if $n \neq k$, then $C_n[\,n=k\,] = 0 = C_k[\,n=k\,]$. In both cases $C_n[\,n=k\,] = C_k[\,n=k\,]$, and therefore

$$T(n,k) = C_k\big([\,n=k\,] - W\big).$$

So the claim holds with $z = [\,n=k\,] - \sum_{j<n} M(n,j)\,z_j \in \mathbb{Z}$; this is the witness given in Lean, where the two cases $n = k$ and $n \neq k$ are closed by `field_simp` (and `ring`). □

Proof of Theorem 1. Let $k \le n$. By Proposition 2, $T(n,k) = z\,C_k$ for some $z \in \mathbb{Z}$, and zC_k is an integer since $C_k \in \mathbb{N}$. This is `result`: `statement`, whose Lean proof gives the witness $z \cdot C_k$ (with `cat k` cast to $\mathbb{Z}$) and checks $T(n,k) = zC_k$ by `push_cast` and `ring`. □

The rest of this section is not formalized.

Corollary 3. *$T(n,k) = C_k \cdot M^{-1}(n,k)$ for all n, k, where M^{-1} is the inverse of the triangle $M(n,k) = [\,k \le n \le 2k\,]$, that is, A111967. Equivalently, $A^{-1} = M^{-1}\,\mathrm{diag}(C_k)$.*

Proof. Let $U(n,k) = T(n,k)/C_k$, so $U(n,k) = 0$ for $k > n$. Multiplying (2) out, using $C_n A(n,j) = M(n,j)$ and $C_n[\,n=k\,] = C_k[\,n=k\,]$, and dividing by C_k, we get for $k \le n$

$$U(n,k) = [\,n=k\,] - \sum_{j<n} M(n,j)\,U(j,k).$$

As M is lower unitriangular, this is exactly the forward-substitution recursion that determines $M^{-1}(n,k)$ for $k \le n$, so $U = M^{-1}$. Equivalently, $A = DM$ with $D = \mathrm{diag}(1/C_n)$, and $A^{-1} = M^{-1}D^{-1}$. □

The witnesses z in the proof of Proposition 2 are therefore the entries of A111967. In an exact computation for $n < 60$ (not part of the proof), every entry of M^{-1} lies in $\{-1, 0, 1\}$, so each $T(n,k)$ in that range is 0 or $\pm C_k$, as in the listed data. We do not claim this in general. The identity of Corollary 3, $\mathrm{A127767}(n,k) = \mathrm{A000108}(k) \cdot \mathrm{A111967}(n,k)$, would be a natural formula line for the entry.

Remark 4 (the sibling triangles)**.** The argument uses nothing about C_n except that it is a nonzero integer. For any sequence $d(n)$ of nonzero integers, the inverse of the triangle $[\,k \le n \le 2k\,]/d(n)$ is $M^{-1}\,\mathrm{diag}(d(k))$, whose (n,k) entry is $d(k)\,\mathrm{A111967}(n,k) \in \mathbb{Z}$. This proves the integrality conjectures of A127710 ($d(n) = n+1$), A127711 ($d(n) = F_{n+1}$), A127755 ($d(n) = n!$) and A127771 ($d(n) = \varphi(n+1)$). It also gives another proof of the integrality in A127793 ($d(n) = \lfloor (n+2)/2 \rfloor$), which Fung and Fried [1] established earlier, and for A127749 ($d(n) = 2n+1$) it gives Mathar's formula. We have not formalized these consequences.

4. Machine verification

Files and declarations. The proof is in `lean/Math0/Frontier/OEIS/A127767.lean` in the math0 repository, `https://github.com/Foundation-0/math0`. It imports the frozen statement file `lean/Math0/Statements/OEIS/A127767.lean` and the audit commands in `lean/Math0/Audit.lean`. It consists of the private theorem `T_eq_int_mul_cat` (Proposition 2), the main declaration `Math0.Frontier.A127767.result : statement`, and the command

```
#math0_settles Math0.Frontier.A127767.result Math0.Statements.A127767.statement
```

From the statement file the proof uses `T_spec` and `cat_pos`. Neither file uses `sorry`, `native_decide`, custom axioms or `set_option`. The proof file uses no `decide`; the only kernel evaluations are the data checks of the statement file, which the proof does not use.

Audit. The command `#math0_settles` (in `lean/Math0/Audit.lean`, for statements that math0 formalizes itself) fails unless all of the following hold:

- `statement` is a `Prop`-valued definition without universe parameters;
- `statement` and `result` depend on no axioms besides `propext`, `Classical.choice` and `Quot.sound`, which rejects `sorryAx`, `Lean.ofReduceBool` (from `native_decide`) and custom axioms;
- the type of `result` is syntactically `statement` or $\neg$ `statement`, up to metadata, with no unfolding.

Here it reports

```
math0 audit: 'Math0.Statements.A127767.statement' depends on axioms [Classical.choice, Quot.sound, propext]
math0 audit: 'Math0.Frontier.A127767.result' depends on axioms [Classical.choice, Quot.sound, propext]
math0 statement check: 'Math0.Frontier.A127767.result' proves 'Math0.Statements.A127767.statement'
```

Toolchain and reproduction. We use Lean 4 [4] `v4.33.1` and Mathlib [3] `v4.33.1`, pinned in `lean/`. The repository also pins formal-conjectures [2] at commit `c77fdd4`; this result does not use it, since its statement is math0's own formalization of the OEIS entry. To reproduce, run `scripts/verify.sh` from the repository root. The script fetches the Mathlib cache, builds all math0 results with `lake build` (which re-runs every audit command), and checks that the pinned `formal_conjectures` and `mathlib` checkouts are unmodified. To check this file alone, run `lake env lean Math0/Frontier/OEIS/A127767.lean` in `lean/`. Once the statement module is built, this takes well under a minute of CPU time on our machine, most of it spent loading Mathlib.

References

[1] S. Fried, *On the inverse of a certain triangular matrix and its connection to the largest odd divisor*, Notes on Number Theory and Discrete Mathematics **32**(2) (2026), 255–262, `https://doi.org/10.7546/nntdm.2026.32.2.255-262`.

[2] M. Firsching, P. Lezeau, S. Mercuri, M. Z. Horváth, Y. Dillies, C. Sönne, E. Wieser, F. Zhang, T. Hubert, B. Agüera y Arcas, and P. Kohli, *Formal Conjectures: An Open and Evolving Benchmark for Verified Discovery in Mathematics*, arXiv:2605.13171, 2026. Repository: `https://github.com/google-deepmind/formal-conjectures`.

[3] The mathlib Community, *The Lean mathematical library*, in: Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020), ACM, 2020, 367–381.

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[5] The OEIS Foundation Inc., *Catalan numbers:* $C(n) = \binom{2n}{n}/(n+1) = (2n)!/(n!(n+1)!)$, entry A000108 in The On-Line Encyclopedia of Integer Sequences, `https://oeis.org/A000108`.

[6] The OEIS Foundation Inc., *Once 1, once 0, repeat, twice 1, twice 0, repeat, thrice 1, thrice 0, ... and so on*, entry A101688 (R. Stephan, 2004) in The On-Line Encyclopedia of Integer Sequences, `https://oeis.org/A101688`.

[7] The OEIS Foundation Inc., *Inverse of number triangle A101688*, entry A111967 (P. Barry, 2005) in The On-Line Encyclopedia of Integer Sequences, `https://oeis.org/A111967`.

[8] The OEIS Foundation Inc., *Inverse of the triangle* $A(n,k) = \text{if}(k \le n, \text{if}(n \le 2k, 1/(n+1), 0), 0)$, entry A127710 (P. Barry, 2007) in The On-Line Encyclopedia of Integer Sequences, `https://oeis.org/A127710`.

[9] The OEIS Foundation Inc., *Inverse of the triangle* $A(n,k) = 1/F(n+1)$ *if* $k \le n \le 2k$, 0 *otherwise*, entry A127711 (P. Barry, 2007) in The On-Line Encyclopedia of Integer Sequences, `https://oeis.org/A127711`.

[10] The OEIS Foundation Inc., *Inverse of number triangle* $A(n,k) = 1/(2n+1)$ *if* $k \le n \le 2k$, 0 *otherwise*, entry A127749 (P. Barry, 2007; formula by R. J. Mathar, April 21, 2021) in The On-Line Encyclopedia of Integer Sequences, `https://oeis.org/A127749`.

[11] The OEIS Foundation Inc., *Inverse of number triangle* $A(n,k) = 1/n!$ *if* $k \le n \le 2k$, 0 *otherwise*, entry A127755 (P. Barry, 2007) in The On-Line Encyclopedia of Integer Sequences, `https://oeis.org/A127755`.

[12] The OEIS Foundation Inc., *Inverse of number triangle* $A(n,k) = 1/C(n)$ *if* $k \le n \le 2k$, 0 *otherwise, where* $C(n) = \text{A000108}(n)$, entry A127767 (P. Barry, January 28, 2007) in The On-Line Encyclopedia of Integer Sequences, `https://oeis.org/A127767`; revision #11, September 30, 2018.

[13] The OEIS Foundation Inc., *Inverse of number triangle* $A(n,k) = 1/\text{Euler_phi}(n+1)$ *if* $k \le n \le 2k$, 0 *otherwise*, entry A127771 (P. Barry, 2007) in The On-Line Encyclopedia of Integer Sequences, `https://oeis.org/A127771`.

[14] The OEIS Foundation Inc., *Inverse of number triangle* $t(n,k) = 1/\lfloor (n+2)/2 \rfloor$ *if* $k \le n \le 2k$, 0 *otherwise*, entry A127793 (P. Barry, 2007; comments by FUNG Cheok Yin, 2022, and S. Fried, 2025) in The On-Line Encyclopedia of Integer Sequences, `https://oeis.org/A127793`.

Produced by Claude (an AI model developed by Anthropic) in the math0 project